

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT II SIGNAL DEGRADATION IN OPTICAL FIBERS

Degree, Semester & Branch: VII Semester B.E.ECE B

Course Code & Title: EC6702 Optical Communication and Networks

Name of the Faculty member: Mrs.R.Ramalakshmi

Name of the Topic: Bending loss and connector loss

Name of the Innovative Practice: Demonstration

Date & Duration: 16.08.2019 & 15 Minutes

ICT Tool Used: Demonstration

Description:

A method demonstration is a teaching method used to communicate an idea with the aid of visuals such as flip charts, posters, power point, etc. A demonstration is the process of teaching someone how to make or do something in a step-by-step process.

The benefits of demonstration as a teaching method are a better learning experience in the classroom for students; the generation interest in the subject; help in developing the spirit of inquiry; students cooperating in the teaching-learning process; and the process of learning becoming permanent in the student's life.

Goals (Learning Outcomes):

- The students will be able to know about the concepts of various losses that occur in the optical fiber communication system.

Use of appropriate methods:

Justification for choosing the topic using demonstration activity:

Demonstration of the experiment shows clear idea about the concepts. The students will be able to gather more information by doing the experiment in the kit rather than explaining the concepts theoretically.

Effective presentation

Implementation(Plan & Execution) with Proof:

The students were asked to come to Optical and Microwave Laboratory after explaining the concepts of bending and connector losses in the class room. Demonstration on bending and connector loss experiment was performed to the students and the students also did the experiment.

Innovative Teaching Method Execution

Demonstration of Bending Loss and Connector Loss



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when the students executed the results during the lab session.
- ✓ During practical lab and in university lab, the students performed well.
- ✓ The students scored good marks in the bending and connector loss topics in the internal test.

Reflective Critique:

• Benefits:

The students were more interested in learning the concepts if we visually show the experiments. They clarified their doubts with the neighbouring students and the faculty. They learnt the concepts of bending loss and connector loss more clearly by performing demonstration. They asked more doubts in the topic and learnt many things.

• Challenges:

Basically some time is needed to explain the basic concepts of bending loss and connector loss. It was somewhat challenging to explain to the slow learners initially, later on after explaining the same through demonstration, it was understood by the students clearly.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2		3					3					2

CO2: The students will be able to analyze the various signal degradation factors associated with optical fiber

References:

1. Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010.
2. John M. Senior, "Optical Fiber communication", Second Edition, Pearson Education, 2007.
3. Ramaswami, Sivarajan and sasaki "Optical Networks ", Morgan kaufmann, 2009
4. <https://en.wikipedia.org/wiki/Demonstration>

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Innovative Practices Description

UNIT III FIBER OPTICAL SOURCES AND COUPLING

Degree, Semester & Branch: VII Semester B.E.ECE B

Course Code & Title: EC6702 Optical Communication and Networks

Name of the Faculty member: Mrs.R.Ramalakshmi

Name of the Topic: Laser Fundamentals

Name of the Innovative Practice: Lecture Video

Date & Duration: 06.09.2019 & 15 Minutes

ICT Tool Used: Lecture Video

Description:

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

Goals (Learning Outcomes):

The students were able to understand the concepts of Laser characteristics clearly by the lecture video.

Use of appropriate methods:

Justification for choosing the topic using demonstration activity:

Lecture video shows clear idea about the concepts. The students will be able to gather more information by visualizing the concepts in the screen rather than explaining the concepts theoretically.

Effective presentation

Implementation(Plan & Execution) with Proof:

A lecture video was played inside the classroom and the students viewed the video.

Innovative Teaching Method Execution
Lecture Video on Laser Fundamentals



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when the students understood the concepts of laser and cleared their doubts.
- ✓ The students performed well in the internal test by scoring good marks in the topic laser.

Reflective Critique:

- **Benefits:**

The students were more interested in learning the concepts if we visually show the videos. They clarified their doubts with the neighbouring students and the faculty. They learnt the concepts of laser fundamentals more clearly by seeing the videos. They asked more doubts in the topic and learnt many things.

- **Challenges:**

Some extra time is required in the class room to show the video.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 3	3									3		

CO3: The students will be able to explain the various optical sources and optical detectors and their use in the optical communication system.

References:

1. Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010.
2. John M. Senior, "Optical Fiber communication", Second Edition, Pearson Education, 2007.
3. Ramaswami, Sivarajan and sasaki "Optical Networks ", Morgan kaufmann, 2009
4. <https://freevideolectures.com/>

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Degree, Semester & Branch: VII Semester B.E.ECE B

Course Code & Title: EC6702 Optical Communication and Networks

Name of the Faculty member: Mrs.R.Ramalakshmi

Name of the Topic: Laser Fundamentals

Name of the Innovative Practice: Lecture Video

Date & Duration: 06.09.2019 & 15 Minutes

ICT Tool Used: Lecture Video

Description:

Over the past few years, videos are being widely used in classrooms for supporting a teacher's curriculum and helping students learn the material faster than ever. Research shows that 94% of the teachers have effectively used videos during the academic year and they have found video learning quite effective, it is even better than teaching students through traditional text-books.

Majority of part of the human brain is devoted towards processing the visual information. Brain responds to visuals fast, better than text or any other kind of learning material. Remembering stuff from the picture is retained in the mind for a longer time. Through videos, students get to process information fast.

Goals (Learning Outcomes):

The students were able to understand the concepts of Laser characteristics clearly by the lecture video.

Use of appropriate methods:

Justification for choosing the topic using demonstration activity:

Lecture video shows clear idea about the concepts. The students will be able to gather more information by visualizing the concepts in the screen rather than explaining the concepts theoretically.

Effective presentation

Implementation(Plan & Execution) with Proof:

A lecture video was played inside the classroom and the students viewed the video.

Innovative Teaching Method Execution
Lecture Video on Laser Fundamentals



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when the students understood the concepts of laser and cleared their doubts.
- ✓ The students performed well in the internal test by scoring good marks in the topic laser.

Reflective Critique:

- **Benefits:**

The students were more interested in learning the concepts if we visually show the videos. They clarified their doubts with the neighbouring students and the faculty. They learnt the concepts of laser fundamentals more clearly by seeing the videos. They asked more doubts in the topic and learnt many things.

- **Challenges:**

Some extra time is required in the class room to show the video.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO 3	3									3		

CO3: The students will be able to explain the various optical sources and optical detectors and their use in the optical communication system.

References:

1. Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010.
2. John M. Senior, "Optical Fiber communication", Second Edition, Pearson Education, 2007.
3. Ramaswami, Sivarajan and sasaki "Optical Networks ", Morgan kaufmann, 2009
4. <https://freevideolectures.com/>

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Innovative Practices Description

UNIT V OPTICAL NETWORKS AND SYSTEM TRANSMISSION

Degree, Semester & Branch: VII Semester B.E.ECE B

Course Code & Title: EC6702 Optical Communication and Networks

Name of the Faculty member: Mrs.R.Ramalakshmi

Name of the Topic: SONET

Name of the Innovative Practice: Flipped class room

Date & Duration: 11.10.2019 & 15 Minutes

ICT Tool Used: Smart Class Room

Description:

A flipped classroom is an instructional strategy and a type of blended learning that reverses the traditional learning environment by delivering instructional content, often online, outside of the classroom. It moves activities, including those that may have traditionally been considered homework, into the classroom.

Learning Outcomes:

The students were able to understand the concepts of SONET/SDH and its frame structure and its usage in practical applications.

Use of appropriate methods:

Justification for choosing the topic using flipped class room activity:

The students were given assignment on the particular topic some few days back and they were asked to discuss the same topic inside the class room. They could be able to deliver more concepts. Since the assignment was given some days back, they got time to prepare and discuss efficiently in the class room.

Effective presentation

Implementation(Plan & Execution) with Proof:

The students discussed the concepts of SONET more effectively in the class room.

Innovative Teaching Method Execution

Flipped Class Room on “ SONET”



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when the students were able to answer well in the class after performing this activity.
- ✓ Most of the students performed well in the internal tests on this particular topic.

Reflective Critique:

- **Benefits:**

The students were more interested in learning the concepts if they discuss themselves. They clarified their doubts with the neighbouring students and the faculty. They learnt the concepts of SONET more clearly by discussing the concepts. They asked more doubts in the topic and learnt many things.

- **Challenges:**

Basically some time is needed to explain the basic concepts of flipped classroom. It was somewhat challenging to explain to the slow learners initially, later on after explaining the same, it was understood by the students clearly.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5		2	2		3					3		

CO5: The students will be able to analyze the Digital Transmission and its associated parameters on system performance.

References:

1. Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010.
2. John M. Senior, "Optical Fiber communication", Second Edition, Pearson Education, 2007.
3. Ramaswami, Sivarajan and sasaki "Optical Networks ", Morgan kaufmann, 2009
4. https://en.wikipedia.org/wiki/Flipped_classroom